Type MS · Areas and perimeters of shapes made from sectors and triangles

Answer 1 of 8 · Question 4 part whole · 7 marks

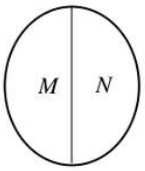
2022 JUN · Paper 02 · extract 1 of 4 · find part whole in this mark scheme extract

4	$O_1A = O_2A = O_1O_2 = 6$ so triangle AO_1O_2 is equilateral	B1
	So angle $AO_1O_2 = \frac{\pi}{3}$	M1
	Area of sector $AO_1BO_2 = \frac{1}{2} \times '\frac{2\pi}{3}' \times 6^2 = (12\pi)$	M1
	Area of triangle $AO_1B = \frac{1}{2} \times 6^2 \times \sin\left('\frac{2\pi}{3}'\right) = 9\sqrt{3}$	M1
	Area of segment $AO_1B = 12\pi - 9\sqrt{3}$	M1
	So $P = 2 \times \left('12\pi - 9\sqrt{3}'\right) = 24\pi - 18\sqrt{3}$ (cm^2)	M1
	Decimal value for reference is 44.221… (cm^2)	A1 (7)
		Total 7 marks

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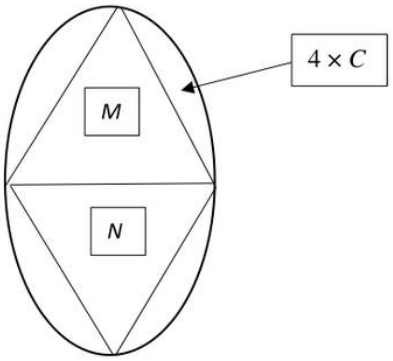
2022 JUN · Paper 02 · extract 2 of 4 · find part whole in this mark scheme extract

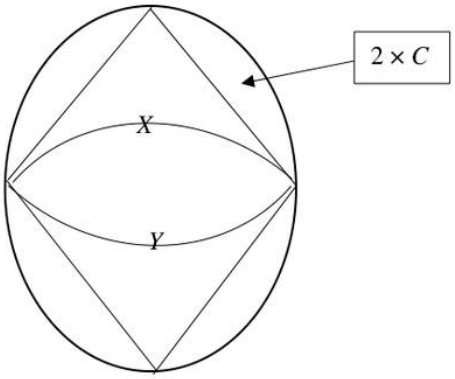
Mark	Notes
B1	For recognising that triangle AO_1O_2 is equilateral. This can be implied by further work. Look at the diagram. The angles are frequently marked on there. **This mark is common to all three methods of finding the area.** **Note**: Correct calculations for either the area of the sector or area of triangle implies the first two marks.
Finds area of segment AO_1B and doubles. (Area M + Area N)  	
M1	For angle $AO_1O_2 = \frac{\pi}{3}$ or 60° **NB** This is often marked on the diagram. Please check that carefully.
M1	Correct method to find area of sector $AO_1BO_2 = \frac{1}{2}\times'\frac{2\pi}{3}'\times 6^2 = (12\pi)$ or $\frac{1}{3}\times\pi\times 6^2 = (12\pi)$
M1	Correct method to find area of triangle $AO_1B = \frac{1}{2}\times 6^2\times\sin\left('\frac{2\pi}{3}'\right) = 9\sqrt{3}$
M1	So area of segment $AO_1B = 12\pi - 9\sqrt{3}$
dM1	For area of $R = 2\times('12\pi - 9\sqrt{3}')$ **Dependent on all 4 previous M marks**
A1	For $24\pi - 18\sqrt{3}$ (cm^2) (Units not required)
ALT	
M1	For angle $AO_1O_2 = \frac{\pi}{3}$ or 60° **NB** This is often marked on the diagram. Please check that carefully.
M1	Correct method to find area of sector $AO_1O_2 = \frac{1}{2}\times'\frac{\pi}{3}'\times 6^2 = (6\pi)$ or $\frac{1}{6}\times\pi\times 6^2$
M1	Correct method to find area of triangle $AO_1O_2 = \frac{1}{2}\times 6^2\times\sin\left('\frac{\pi}{3}'\right) = 9\sqrt{3}$ OR height of triangle $AO_1O_2 = 3\sqrt{3} \Rightarrow$ Area $AO_1O_2 = \frac{1}{2}\times 3\sqrt{3}\times 6 = 9\sqrt{3}$
M1	So area of segment C (on diagram) $= '6\pi' - '9\sqrt{3}'$

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Method 1 - 2 triangles + 4 segments = $M+N+4\times C$	
	
dM1	Area of $R = 2\times\Delta AO_1O_2 + 4\times$segment $AO_1O_2 = 2\times\left('9\sqrt{3}'\right)+4\left('6\pi'-'9\sqrt{3}'\right)$ **Dependent on all 4 previous M marks**
A1	For $24\pi-18\sqrt{3}$ $\left(\text{cm}^2\right)$ (Units not required)

Method 2 – 2 sectors + 2 segments = $X+Y+2C$	
	
dM1	Area of $R = 2\times$ Sector $AO_1O_2 + 2\times$segment $AO_1O_2 = 2\times'6\pi'+2\left('6\pi'-'9\sqrt{3}'\right)$ **Dependent on all 4 previous M marks**
A1	For $24\pi-18\sqrt{3}$ $\left(\text{cm}^2\right)$ (Units not required)

Type MS · Areas and perimeters of shapes made from sectors and triangles

Answer 1 of 8 · Question 4 part whole · 7 marks

2022 JUN · Paper 02 · extract 4 of 4 · find part whole in this mark scheme extract

Question number	Scheme	Marks
	$b \quad -(6+2p)$	

Type MS · Areas and perimeters of shapes made from sectors and triangles

Answer 2 of 8 · Question 3 part c · 2 marks

2022 JUN · Paper 01 (regional R) · extract 1 of 2 · find part c in this mark scheme extract

3 (a)	$(\frac{1}{2} \times 4^2 \times \theta = 2\pi)$ $\frac{1}{2} \times$ " 16" $\times \theta = 2\pi \Rightarrow \theta = ...$ $\theta = \frac{2\pi}{8} = \frac{\pi}{4}$ oe	M1 A1 [2]
(b)	(arc SR =) $4 \times$ " $\frac{\pi}{4}$ " $(= \pi)$ $(4 + 4 + \pi) = 11.1$ (cm) Accept awrt 11.1 (cm)	M1 A1ft [2]
(c)	$\frac{1}{2} \times (1) \times 4 \times \sin$ " $\frac{\pi}{4}$ " $(= \sqrt{2})$ $2\pi - \sqrt{2}$ oe	M1 A1 [2]
		Total 6 marks

Part	Mark	Additional Guidance
(a)	M1	Use of the correct formula for the area of sector, with correct substitution of 4 and 2π, an attempt to square r and to rearrange to find a value for θ. The rearrangement has to be correct for this mark.
	A1	Accept any valid exact equivalent. Do not accept answers in degrees or answers not given exactly for A1. However, 45° or 0.785(3981…) stated with no method - award implied M1.
(b)	M1	Use of correct formula for length of an arc with the correct radius and "their θ" (allow working in degrees for this mark) from part (a). Do **not** allow $\theta = \frac{\pi}{2}$ or 90°
	A1ft	Allow **ft** of their angle from part (a), if complete method shown and if their answer is correct to 3sf. **Note this is ft mark. Check working carefully.**
(c)	M1	Use of correct formula for area of a triangle with correct sides substituted and "their θ" (allow working in degrees) from part (a). Do **not** allow $\theta = \frac{\pi}{2}$ or 90°
	A1	For the correct answer shown.

Type MS · Areas and perimeters of shapes made from sectors and triangles

Answer 2 of 8 · Question 3 part c · 2 marks

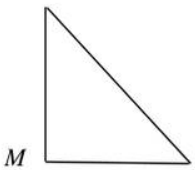
2022 JUN · Paper 01 (regional R) · extract 2 of 2 · find part c in this mark scheme extract

Question number	Scheme	Marks

Type MS · Areas and perimeters of shapes made from sectors and triangles

Answer 3 of 8 · Question 9 part a · 7 marks

2023 NOV · Paper 02 · extract 1 of 3 · find part a in this mark scheme extract

Question	Scheme	Marks
9(a)	$\angle AOB = \dfrac{1.8x}{x} = 1.8$	B1
	O  M B $MB = x\sin 0.9 \Rightarrow AB = 2x\sin 0.9$	M1
	Hence diameter $AC = 2x\sin 0.9 + 2x$ [or Radius = $x\sin 0.9 + x$]	dM1A1
	Arc length of semicircle: $\dfrac{\pi D}{2} = \dfrac{\pi\left(2x\sin 0.9 + 2x\right)}{2} = \pi x\left(1+\sin 0.9\right)$	M1
	Perimeter of logo: $P = \pi x\left(1+\sin 0.9\right) + 2x + 1.8x \Rightarrow P = x\left(\pi + \pi\sin 0.9 + 3.8\right)$ [a=1, b =3.8]	M1A1 [7]
(b)	$D = 2\times 10\times\sin 0.9 + 2\times 10 = 35.6665...$	M1
	Area of semicircle $= \dfrac{\pi\times 35.6665...^2}{4} = 499.553...$	M1
	Area of sector $= 1.8\times\dfrac{10^2}{2} = 90$	B1
	Area of segment $= 90 - \dfrac{1}{2}\times 10\times 10\times\sin 1.8 = 41.307...$ OR Area of triangle $= \dfrac{1}{2}\times 10\times 10\times\sin 1.8 = 48.692$	B1
	Area of logo $= 499.553 - 41.307 = 458.245... \approx$ awrt $458\left(\text{cm}^2\right)$	M1A1 [6]
		Total 13 marks

Part	Mark	Notes
(a)	B1	For finding angle AOB = 1.8 in radians only seen anywhere.
	M1	For find the length AB or $\dfrac{1}{2}AB$ using their angle.

Type MS · Areas and perimeters of shapes made from sectors and triangles

Answer 3 of 8 · Question 9 part a · 7 marks

2023 NOV · Paper 02 · extract 2 of 3 · find part a in this mark scheme extract

		$AB = 2x\sin 0.9$ $\frac{1}{2}AB = x\sin 0.9$ [Allow $AB = 1.57x$] Please check their diagram for their labelling/notation.
	dM1	For attempting to find the diameter (or radius) of the semi-circle ADC Allow $2x + 1.57x$ or $3.57x$ [Dependent on the previous M mark]
	A1	For the correct expression for the diameter $\left[2x\sin 0.9 + 2x\right]$ or the radius $\left[x\sin 0.9 + x\right]$ of the semi-circle.
	M1	For finding the arc length of the semi-circle. This can be simplified or unsimplified. [Allow also $1.79\pi x$]
	M1	For finding the perimeter of the logo using their values **in terms of x and sin 0.9 only** That is: $P = 1.8x + 2x +$ their arc length of ADC
	A1	For the correct expression exactly as written.
	ALT – Uses cosine rule	
	B1	For finding angle $AOB = 1.8$
	M1	Finds the length AB or $\frac{1}{2}AB$ using their angle. [They must sqrt to find AB] $AB = \sqrt{x^2 + x^2 - 2\times x\times x\cos 1.8}$
	dM1	For attempting to find the diameter (or radius) of the semi-circle ADC $AC = 2x + \text{'}x\sqrt{2-2\cos 1.8}\text{'}$
	A1	For the correct diameter or radius of the circle.
	M1	For finding the arc length of the semi-circle. $P = \frac{\pi\left(x\sqrt{2-2\cos 1.8}\right)}{2}$
	M1	For finding the perimeter of the logo using their values. That is: $P = 1.8x + 2x + \text{'}\pi x\left(1+\sin 0.9\right)\text{'}$ They must change cosine to sine for this mark. $\frac{1}{2}\left(\sqrt{2-2\cos 1.8}\right) = \sin 0.9$ must be explicitly seen **NOTE:** $\left[\sin\theta = \sqrt{\frac{1-\cos 2\theta}{2}}\right]$
	A1	For the correct expression exactly as written.
(b)	**Allow work in degrees in this part as it makes no difference.**	
	M1	For attempting to find the diameter or radius of the semicircle. Accept: $D = 2\times 10\times\sin\angle\text{their } AOB + 2\times 10 = \left(35.6665...\right)$ $r = \frac{2\times 10\times\sin\angle\text{their } AOB + 2\times 10}{2} = \left(17.833...\right)$
	M1	For using their D ,[where $D \neq 10$] to find the area of the **semicircle**. $\text{Area}_{semicircle} = \frac{\pi\times\left(\text{'}35.6665...\text{'}\right)^2}{4}$ OR $\frac{\pi\times 17.8333...^2}{2} = \left(499.553...\right)$ **Note: This is an A mark in Epen**
	B1	For finding the area of the sector If they work in degrees, accept awrt 90. **This is not a ft mark.**

Type MS · Areas and perimeters of shapes made from sectors and triangles

Answer 3 of 8 · Question 9 part a · 7 marks

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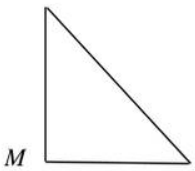
	B1	For finding the area of the segment, awrt 41 OR for finding the area of the triangle $OAB = \frac{1}{2} \times 10^2 \times \sin 1.8 = [48.692...]$ **This is not a ft mark.**
	M1	For finding the area of the logo Area = Their area of semicircle – their area of the segment. Or Area = Area of whole shape – area of sector
	A1	For the correct value of the area of the logo.

Question	Scheme	Marks

Type MS · Areas and perimeters of shapes made from sectors and triangles

Answer 4 of 8 · Question 9 part b · 6 marks

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Question	Scheme	Marks
9(a)	$\angle AOB = \frac{1.8x}{x} = 1.8$	B1
	 $MB = x\sin 0.9 \Rightarrow AB = 2x\sin 0.9$	M1
	Hence diameter $AC = 2x\sin 0.9 + 2x$ [or Radius = $x\sin 0.9 + x$]	dM1A1
	Arc length of semicircle: $\frac{\pi D}{2} = \frac{\pi(2x\sin 0.9 + 2x)}{2} = \pi x(1+\sin 0.9)$	M1
	Perimeter of logo: $P = \pi x(1+\sin 0.9) + 2x + 1.8x \Rightarrow P = x(\pi + \pi\sin 0.9 + 3.8)$ [a=1, b =3.8]	M1A1 [7]
(b)	$D = 2\times 10\times\sin 0.9 + 2\times 10 = 35.6665...$	M1
	Area of semicircle $= \frac{\pi\times 35.6665...^2}{4} = 499.553...$	M1
	Area of sector $= 1.8\times\frac{10^2}{2} = 90$	B1
	Area of segment $= 90 - \frac{1}{2}\times 10\times 10\times\sin 1.8 = 41.307...$ OR Area of triangle $= \frac{1}{2}\times 10\times 10\times\sin 1.8 = 48.692$	B1
	Area of logo $= 499.553 - 41.307 = 458.245... \approx$ awrt $458\,(\text{cm}^2)$	M1A1 [6]
		Total 13 marks

Part	Mark	Notes
(a)	B1	For finding angle AOB = 1.8 in radians only seen anywhere.
	M1	For find the length AB or $\frac{1}{2}AB$ using their angle.

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Answer 4 of 8 · Question 9 part b · 6 marks

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		$AB = 2x\sin 0.9$ $\frac{1}{2}AB = x\sin 0.9$ [Allow $AB = 1.57x$] Please check their diagram for their labelling/notation.
	dM1	For attempting to find the diameter (or radius) of the semi-circle ADC Allow $2x + 1.57x$ or $3.57x$ [Dependent on the previous M mark]
	A1	For the correct expression for the diameter $\left[2x\sin 0.9 + 2x\right]$ or the radius $\left[x\sin 0.9 + x\right]$ of the semi-circle.
	M1	For finding the arc length of the semi-circle. This can be simplified or unsimplified. [Allow also $1.79\pi x$]
	M1	For finding the perimeter of the logo using their values **in terms of x and sin 0.9 only** That is: $P = 1.8x + 2x +$ their arc length of ADC
	A1	For the correct expression exactly as written.
	ALT – Uses cosine rule	
	B1	For finding angle $AOB = 1.8$
	M1	Finds the length AB or $\frac{1}{2}AB$ using their angle. [They must sqrt to find AB] $AB = \sqrt{x^2 + x^2 - 2\times x\times x\cos 1.8}$
	dM1	For attempting to find the diameter (or radius) of the semi-circle ADC $AC = 2x + \text{'}x\sqrt{2-2\cos 1.8}\text{'}$
	A1	For the correct diameter or radius of the circle.
	M1	For finding the arc length of the semi-circle. $P = \frac{\pi\left(x\sqrt{2-2\cos 1.8}\right)}{2}$
	M1	For finding the perimeter of the logo using their values. That is: $P = 1.8x + 2x + \text{'}\pi x\left(1+\sin 0.9\right)\text{'}$ They must change cosine to sine for this mark. $\frac{1}{2}\left(\sqrt{2-2\cos 1.8}\right) = \sin 0.9$ must be explicitly seen **NOTE:** $\left[\sin\theta = \sqrt{\frac{1-\cos 2\theta}{2}}\right]$
	A1	For the correct expression exactly as written.
(b)	**Allow work in degrees in this part as it makes no difference.**	
	M1	For attempting to find the diameter or radius of the semicircle. Accept: $D = 2\times 10\times\sin\angle\text{their } AOB + 2\times 10 = \left(35.6665...\right)$ $r = \frac{2\times 10\times\sin\angle\text{their } AOB + 2\times 10}{2} = \left(17.833...\right)$
	M1	For using their D ,[where $D \neq 10$] to find the area of the **semicircle**. $\text{Area}_{semicircle} = \frac{\pi\times\left(\text{'}35.6665...\text{'}\right)^2}{4}$ OR $\frac{\pi\times 17.8333...^2}{2} = \left(499.553...\right)$ **Note: This is an A mark in Epen**
	B1	For finding the area of the sector If they work in degrees, accept awrt 90. **This is not a ft mark.**

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Answer 4 of 8 · Question 9 part b · 6 marks

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	B1	For finding the area of the segment, awrt 41 OR for finding the area of the triangle $OAB = \frac{1}{2} \times 10^2 \times \sin 1.8 = [48.692...]$ **This is not a ft mark.**
	M1	For finding the area of the logo Area = Their area of semicircle – their area of the segment. Or Area = Area of whole shape – area of sector
	A1	For the correct value of the area of the logo.

Question	Scheme	Marks

Type MS · Areas and perimeters of shapes made from sectors and triangles

Answer 5 of 8 · Question 3 part b · 4 marks

2025 JUN · Paper 01 · extract 1 of 2 · find part b in this mark scheme extract

3 (a)	$2\times7\times\theta=\frac{28\pi}{9}$ oe or $2\times\pi\times7\times\frac{angle}{360}=\frac{14\pi}{9}$ oe **and** attempts to change back to radians $\theta=\frac{2\pi}{9}$	M1 A1 [2]
(b)	Area of sector $A=\frac{1}{2}\times7^2\times'\frac{2\pi}{9}'$ Area of triangle $A=\frac{1}{2}\times7^2\times\sin'\frac{2\pi}{9}'$ Area of segment $[A=][2\times]\left(\frac{1}{2}\times7^2\times'\frac{2\pi}{9}'-\frac{1}{2}\times7^2\times\sin'\frac{2\pi}{9}'\right)=2.71$ (awrt)	M1 M1 ddM1A1 [4]
		Total marks 6

Part	Mark	Additional Guidance
(a)	M1	For forming a correct equation using arc length and $r=7$ If attempted by degrees, must see attempts to change degrees to radians
	A1	For $\theta=\frac{2\pi}{9}$ oee (or exact equivalent)
(b)	M1	For a correct method finding area of the sector using **their** θ $A=\frac{1}{2}\times7^2\times'\frac{2\pi}{9}'$ or $A=\pi\times7^2\times\frac{'40'}{360}$ $\left(\text{Note}:\frac{2\pi}{9}=40°\right)$
	M1	For a fully correct method finding the area of a triangle using their θ, can be implied by the correct area of triangle 15.7 awrt
	ddM1	For finding the difference between the area of the sector and the area of the triangle. The subtraction must be shown, or it could be implied by correct final answer. Dependent on **both** previous M marks
	A1	For 2.71 (awrt) ignore unit

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Answer 5 of 8 · Question 3 part b · 4 marks

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Question number	Scheme	Marks

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Answer 6 of 8 · Question 3 part whole · 6 marks

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Number		
3	$(l = r\theta =)6\theta$ or $(l = \frac{\theta}{360} \times 2\pi r =) \frac{\theta}{360} \times 2\pi \times 6$ oe	M1
	$6+6+"6\theta" = 12+\pi$ or $6+6+"\frac{\theta}{360} \times 2\pi \times 6" = 12+\pi$ oe	M1
	$(6\theta = \pi) \Rightarrow \theta = \frac{\pi}{6}$ or $\theta = 30$	A1
	$\left(A = \frac{1}{2}r^2\theta - \frac{1}{2}r^2\sin\theta =\right) \frac{1}{2}6^2 "\frac{\pi}{6}" - \frac{1}{2}6^2 \sin"\frac{\pi}{6}"$	M1 M1
	or	
	$\frac{"30"}{360} \times \pi \times 6^2 - \frac{1}{2}6^2 \sin"30"$	A1
	$= 3\pi - 9$ oe	[6]
		Total 6 marks

Type MS · Areas and perimeters of shapes made from sectors and triangles

Answer 6 of 8 · Question 3 part whole · 6 marks

2025 JUN · Paper 02 (regional R) · extract 2 of 8 · find part whole in this mark scheme extract

Part	Mark	Additional Guidance
	M1	For a correct vector path for at least one of $\overrightarrow{BC}$ or $\overrightarrow{DC}$ or $\overrightarrow{CB}$ or $\overrightarrow{CD}$ Eg $\overrightarrow{BC} = \overrightarrow{BA} + \overrightarrow{AC}$ or $\overrightarrow{DC} = \overrightarrow{DA} + \overrightarrow{AC}$ This mark may be implied by the next method mark – ie by correctly adding vectors in terms of **a** and **b**. Accept poor notation Eg for $\overrightarrow{BC}$ accept BC or $\lvert \overrightarrow{BC} \rvert$ or $\lvert BC \rvert$
	M1	For a correct unsimplified vector in terms of **a** and **b** for $\overrightarrow{BC}$ or $\overrightarrow{CB}$ or $\overrightarrow{DC}$ or $\overrightarrow{CD}$ Don't accept $\overrightarrow{BC} = +3\mathbf{a} + 2\mathbf{b} - 5\mathbf{a} + 3\mathbf{b}$ (as this is $\overrightarrow{CB}$) **Don't accept unlabelled vectors** Accept poor notation Eg for $\overrightarrow{BC}$ accept BC or $\lvert \overrightarrow{BC} \rvert$ or $\lvert BC \rvert$
	A1	For a correct simplified vector in terms of **a** and **b** for $\overrightarrow{BC}$ or $\overrightarrow{CB}$ or $\overrightarrow{DC}$ or $\overrightarrow{CD}$ In order to gain this accuracy mark we must see the first **or** second M mark (ie the adding together of the paths in vector notation or in terms of the vectors in **a** and **b**) **Do not allow poor notation.** **A correct simplified vector alone does not imply previous marks.**
	A1 cso*	For a correct simplified vector in terms of **a** and **b** for $\overrightarrow{BC}$ or $\overrightarrow{CB}$ or $\overrightarrow{DC}$ or $\overrightarrow{CD}$ and a clear (brief) conclusion. Eg $\overrightarrow{BC} = -\overrightarrow{DA}$ $\overrightarrow{BC} = \overrightarrow{AD}$ $\overrightarrow{DC} = -\overrightarrow{BA}$ $\overrightarrow{DC} = \overrightarrow{AB}$ shown or # * or Candidates may also work with 2 correctly paired vectors and **state $DC = AB$ and $BC = AD$** or $\overrightarrow{DC}$ **is parallel to** $\overrightarrow{AB}$ **and** $\overrightarrow{BC}$ **is parallel to** $\overrightarrow{AD}$ to gain this mark, with also a brief conclusion, if all their work is correct for the vectors involved. Minimum steps shown, **ignore any incorrect vectors** that they are not using to draw their conclusion.

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Answer 6 of 8 · Question 3 part whole · 6 marks

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Question number	Scheme	Marks
2 (a)	$(-3,9)$	B1
	$\left(m_{PQ}=\right)\dfrac{18-3}{3--7}\left(=\dfrac{3}{2}\right)$ oe	M1
	$\left(m_t=\right)=-\dfrac{1}{\text{"}\frac{18-3}{3--7}\text{"}}=\left(-\dfrac{1}{\text{"}\frac{3}{2}\text{"}}\right)\left(=-\dfrac{2}{3}\right)$ oe	M1
	$y-\text{"9"}=\text{"}-\dfrac{2}{3}\text{"}(x+\text{"3"})$	M1
	$3y+2x-21=0$ oe eg $-6y-4x+42=0$	A1 [5]
(b)	$3\times0+2x-21=0\Rightarrow x=(10.5)$	M1
	$(QR^2=)(3-\text{"}-3\text{"})^2+(18-\text{"9"})^2\left(=117\right)$ or $(QR=)\sqrt{(3-\text{"}-3\text{"})^2+(18-\text{"9"})^2}\left(=3\sqrt{13}\right)$	M1
	$(RS^2=)(\text{"10.5"}-\text{"}-3\text{"})^2+(0-\text{"9"})^2\left(=\dfrac{1054}{4}\right)$	M1
	or $(RS=)\sqrt{(\text{"10.5"}-\text{"}-3\text{"})^2+(0-\text{"9"})^2}\left(=\dfrac{9\sqrt{13}}{2}\right)$	
	$\dfrac{1}{2}\times\text{"}3\sqrt{13}\text{"}\times\text{"}\dfrac{9\sqrt{13}}{2}\text{"}$	ddM1
	$\dfrac{351}{4}$ oe (87.75)	A1 [5]
ALT1	$3\times0+2x-21=0\Rightarrow x=(10.5)$	M1
	$\left(\dfrac{1}{2}\right)\begin{vmatrix}3 & \text{"}-3\text{"} & \text{"10.5"} & 3\\ 18 & \text{"9"} & 0 & 18\end{vmatrix}$ oe eg $\left(\dfrac{1}{2}\right)\begin{vmatrix}\text{"10.5"} & \text{"}-3\text{"} & 3 & \text{"10.5"}\\ 0 & \text{"9"} & 18 & 0\end{vmatrix}$ oe	M1 M1
	$\dfrac{1}{2}\left(\left(3\times9+-3\times0+10.5\times18\right)-\left(3\times0+10.5\times9+-3\times8\right)\right)$ oe	ddM1
	$\dfrac{351}{4}$ oe (87.75)	A1
	Finding the area of a rectangle and subtracting area of triangles	
ALT2a	$3\times0+2x-21=0\Rightarrow x=0$	M1
	$(\text{"10.5"}-\text{"}-3\text{"})\times18=243$	M1
	$\dfrac{1}{2}\times(18-\text{"9"})\times(3-\text{"}-3\text{"})\left(=27\right)$	
	$\dfrac{1}{2}\times(\text{"10.5"}-3)\times(18(-0))\left(=\dfrac{135}{2}\right)$	M1 M1
	$\dfrac{1}{2}\times(\text{"10.5"}-\text{"}-3\text{"})\times(\text{"9"}(-0))\left(=\dfrac{243}{4}\right)$	
	$243-\text{"27"}-\text{"}\dfrac{135}{2}\text{"}-\text{"}\dfrac{243}{4}\text{"}$	ddM1
	$\dfrac{351}{4}$ oe (87.75)	A1

Type MS · Areas and perimeters of shapes made from sectors and triangles

Answer 6 of 8 · Question 3 part whole · 6 marks

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ALT2b	Let M be the intersection of PQ with the x-axis	M1
	$3\times 0+2x-21=0\left(\Rightarrow x=10.5\right)$	M1
	$\left(y=\frac{3}{2}x+13.5\Rightarrow\ x=\right)-9$	M1
	$\left(\text{Area of triangle } SQM\ =\right)\frac{1}{2}\times\left("10.5"-"-9"\right)\times 18\left(=\frac{351}{2}\right)$	M1
	$\left(\text{Area of triangle } SRM\ =\right)\frac{1}{2}\times\left("10.5"-"-9"\right)\times 9\left(=\frac{351}{4}\right)$	ddM1
	$"\frac{351}{2}"-"\frac{351}{4}"$	A1
	$\frac{351}{4}$ oe (87.75)	
ALT3	Usually with a good diagram, the candidate can work out that triangles SRM and QRS are congruent (SAS) If spotting this, this is full marks or 0 marks (because without the correct coordinates, this is not so)	
	$\frac{1}{2}\times 19.5\times 9=\frac{351}{4}$	M1M1 M1M1 ddM1A1
		Total 10 marks

Type MS · Areas and perimeters of shapes made from sectors and triangles

Answer 6 of 8 · Question 3 part whole · 6 marks

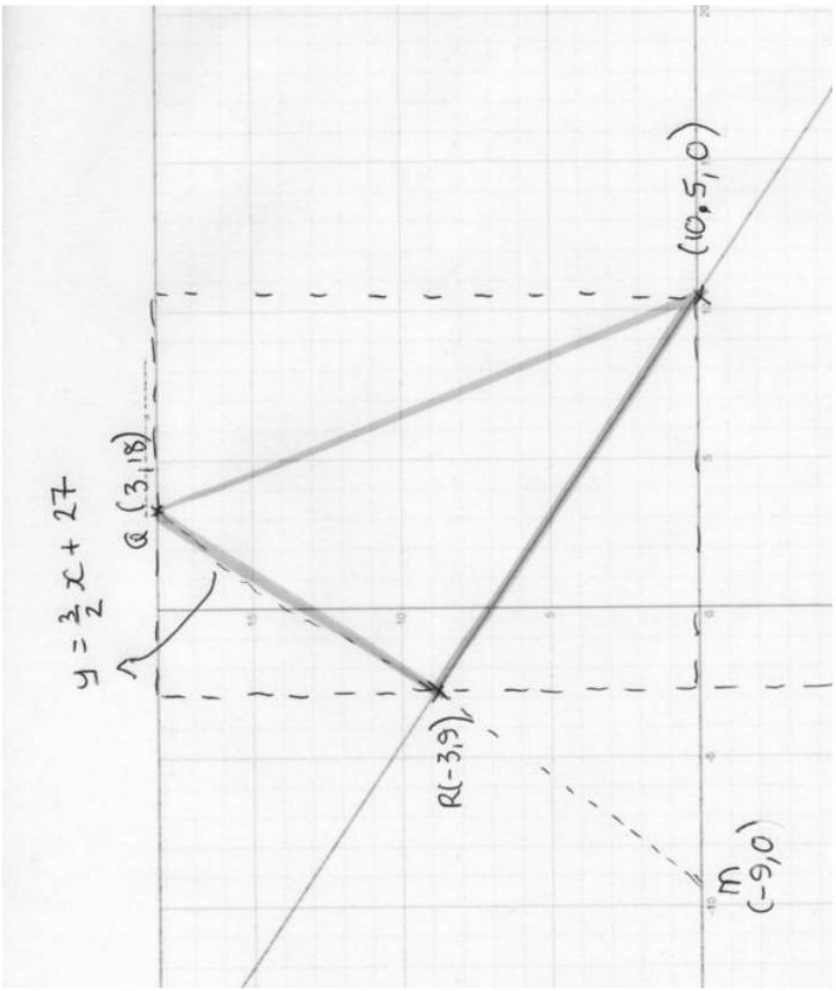
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Part	Mark	Additional Guidance
(a)	B1	For (–3,9)
	M1	For the method for gradient of PQ in unsimplified form
	M1	For finding the negative reciprocal of their gradient of PQ, does not need to be simplified. This mark is not dependent, so may be awarded for finding the negative reciprocal of any gradient believed to be their gradient of PQ
	M1	For a full and correct method to find the equation of the line. Doesn't need to be simplified. Ft their grad for l, so long as it involves a changed gradient from PQ and ft their point R. If using $y = mx + c$, a full method includes correct rearrangement to find c. If the equation is not the correct one and they're working with their values, this rearrangement must be shown or check their c is correct for their values. $c = 7$ implies this mark. Full equation needs to be seen.
	A1	For the equation shown oe integer coefficients.
(b)	M1	For substituting $y = 0$ into their equation for l and correctly solving to find a value of x This mark may be implied by $(x =)$ 10.5
	M1	For a correct method to find QR or QR^2 using their coordinates for point R
	M1	For a correct method to find RS or RS^2 using their coordinates for point R and their value for the x coordinate of S
	ddM1	For a correct method to find the area of the triangle. Dependent on both previous method marks.
	A1	Correct answer as shown.
ALT1	M1	For substituting $y = 0$ into their equation for l and solving to find a value of x This mark may be implied by $(x =)$ 10.5
	M1	For a correct matrix array, condone round or brackets, using their coordinates for R and their 10.5, the 0.5 doesn't need to be present. Be aware of all the equivalents, including reverse order or 'upside down'. May be implied by correct working
	M1	For a correct determinant array, using their coordinates for R and their 10.5, with 0.5 outside. May be implied correct working.
	ddM1	For a correct calculation of the determinant do need to see multiplication by 0.5. May be implied by a correct answer. If the candidate is using their values rather than the correct values, they must show the products (or at the minimum each product evaluated) and subtraction correctly. Dependent on both previous method marks.
	A1	Correct answer as shown.
ALT2a	M1	For substituting $y = 0$ into their equation for l and solving to find a value of x This mark may be implied by $(x =)$ 10.5
	M1	For a correct method to find the area of the rectangle, using their R and their 10.5.
	M1	For a correct method to find the area of the surrounding triangles
	ddM1	For a correct method to find the area of triangle QRS
	A1	Correct answer as shown.
ALT2b	M1	For substituting $y = 0$ into their equation for l and solving to find a value of x This mark may be implied by $(x =)$ 10.5
	M1	For – 9 This is the intersection of PQ with the x-axis The mark is for the number **not** the method
	M1	For a correct method to find the area of triangle SQM
	ddM1	For a correct method to find the area of triangle SPM
	A1	Correct answer as shown.

Type MS · Areas and perimeters of shapes made from sectors and triangles

Answer 6 of 8 · Question 3 part whole · 6 marks

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Type MS · Areas and perimeters of shapes made from sectors and triangles

Answer 6 of 8 · Question 3 part whole · 6 marks

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Question number	Scheme	Mark
3	$(l = r\theta =)6\theta$ or $(l = \frac{\theta}{360} \times 2\pi r =) \frac{\theta}{360} \times 2\pi \times 6$ oe	M1
	$6+6+"6\theta" = 12+\pi$ or $6+6+"\frac{\theta}{360} \times 2\pi \times 6" = 12+\pi$ oe	M1
	$(6\theta = \pi) \Rightarrow \theta = \frac{\pi}{6}$ or $\theta = 30$	A1
	$\left(A = \frac{1}{2}r^2\theta - \frac{1}{2}r^2 \sin\theta =\right) \frac{1}{2}6^2 "\frac{\pi}{6}" - \frac{1}{2}6^2 \sin"\frac{\pi}{6}"$	M1 M1
	or $\frac{"30"}{360} \times \pi \times 6^2 - \frac{1}{2}6^2 \sin"30"$ $= 3\pi - 9$ oe	A1 [6]
		Total 6 marks

Part	Mark	Additional Guidance
Students may work in radians or degrees. If using both, marks may be given for accurate use of formulas.		
	M1	Correct use of formula for length of an arc.
	M1	Allow use of their length of arc, ie for following through their length of arc and equating an equation of the form $6+6+a\theta = 12+\pi$ or $6+6+\frac{\theta}{360} \times 2\pi \times a = 12+\pi$ a doesn't have to be an integer, can be irrational. This may be done in stages. Take care to look out for equivalences, where stages have been missed, this is not a show that question. For example, $"6\theta" = \pi$ would imply both method marks as the candidate could have realised the 12 cancels from each side. As can often be the case, a correct angle will imply method marks (unless incorrect method shown). General marking principle.
	A1	$\theta = \frac{\pi}{6}$
	M1	Correct use and substitution into the formula for area of sector using their θ
	M1	Correct use and substitution into the of formula for area of triangle, using their θ
	A1	$3\pi - 9$ oe (correct decimal is 0.4247…. and is likely to imply correct method)

Type MS · Areas and perimeters of shapes made from sectors and triangles

Answer 6 of 8 · Question 3 part whole · 6 marks

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Question number	Scheme	Marks

Type MS · Areas and perimeters of shapes made from sectors and triangles

Answer 7 of 8 · Question 4 part a · 4 marks

2022 JAN · Paper 01 (regional R) · extract 1 of 2 · find part a in this mark scheme extract

4 (a)	$\frac{1}{2}.\frac{\pi}{3}r^2-\frac{1}{2}.r.\frac{3r}{4}\sin\left(\frac{\pi}{3}\right)(=79.5)$	M1 M1
	$\frac{\pi}{6}r^2-\frac{3\sqrt{3}}{16}r^2=79.5$ leading to $r=$	ddM1
	$r=\sqrt{\frac{79.5}{\left(\frac{\pi}{6}-\frac{3\sqrt{3}}{16}\right)}}=20$	A1 [4]
(b)	$(BD^2)="20"^2+\left(\frac{3}{4}\times"20"\right)^2-2\times20\times\left(\frac{3}{4}\times"20"\right)\cos\frac{\pi}{3}$	M1
	$=5\sqrt{13}$ or awrt 18.02	M1
	Perimeter $=("20"-15)+\frac{"20"\pi}{3}+"5\sqrt{13}"=44$	M1 A1 [4]
		Total 8 marks

Type MS · Areas and perimeters of shapes made from sectors and triangles

Answer 7 of 8 · Question 4 part a · 4 marks

2022 JAN · Paper 01 (regional R) · extract 2 of 2 · find part a in this mark scheme extract

Question number	Scheme	Marks

Type MS · Areas and perimeters of shapes made from sectors and triangles

Answer 8 of 8 · Question 4 part b · 4 marks

2022 JAN · Paper 01 (regional R) · extract 1 of 2 · find part b in this mark scheme extract

4 (a)	$\frac{1}{2}.\frac{\pi}{3}r^2-\frac{1}{2}.r.\frac{3r}{4}\sin\left(\frac{\pi}{3}\right)(=79.5)$	M1 M1
	$\frac{\pi}{6}r^2-\frac{3\sqrt{3}}{16}r^2=79.5$ leading to $r=$	ddM1
	$r=\sqrt{\frac{79.5}{\left(\frac{\pi}{6}-\frac{3\sqrt{3}}{16}\right)}}=20$	A1 [4]
(b)	$(BD^2)="20"^2+\left(\frac{3}{4}\times"20"\right)^2-2\times20\times\left(\frac{3}{4}\times"20"\right)\cos\frac{\pi}{3}$	M1
	$=5\sqrt{13}$ or awrt 18.02	M1
	Perimeter $=("20"-15)+\frac{"20"\pi}{3}+"5\sqrt{13}"=44$	M1 A1 [4]
		Total 8 marks

Type MS · Areas and perimeters of shapes made from sectors and triangles

Answer 8 of 8 · Question 4 part b · 4 marks

2022 JAN · Paper 01 (regional R) · extract 2 of 2 · find part b in this mark scheme extract

Question number	Scheme	Marks